

Chapter 5

States of Consciousness



States of Consciousness

- Consciousness: your sensations and perceptions of external events and your self-awareness of mental events including thoughts, memories, and feelings about your experiences and yourself
- Waking consciousness: normal, clear alert awareness
- Altered state of consciousness (ASC): changes that occur in quality and pattern of mental activity; different from waking consciousness



Need for Sleep

- Sleep is an innate *biological rhythm* that can never be entirely ignored
- Repair/restorative theories of sleep: lowering body & brain activity & metabolism during sleep may help conserve energy and lengthen life
- Sleep deprivation: sleep loss; with few exceptions, 4 days or more without sleep becomes hell for everyone



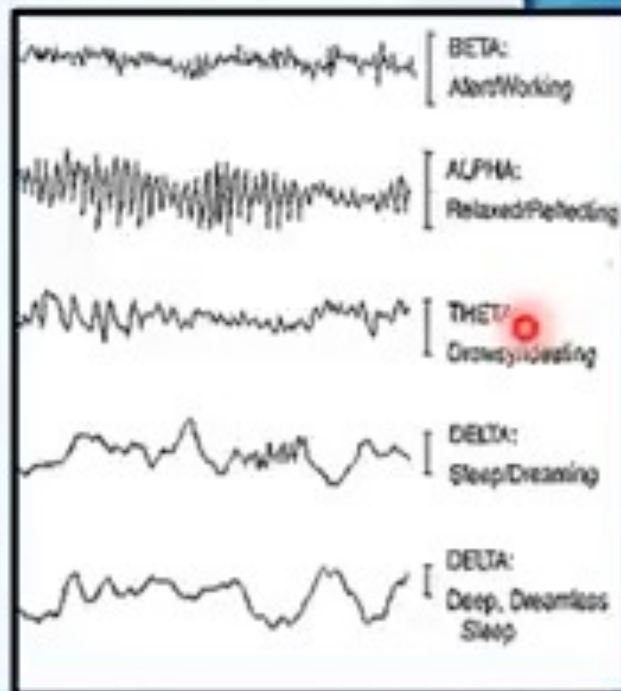
Need for Sleep (cont'd)

- Hypersomnia: excessive daytime sleepiness; arises after even a few hours of sleep loss
- A day or more of sleep deprivation can lead to difficulty staying alert, *microsleeps*, and even *sleep-deprivation psychosis*
 - Microsleep: brief shift in brain activity to normal pattern normally recorded during sleep
 - Sleep-deprivation psychosis: confusion, disorientation, delusions, and hallucinations that occur because of sleep loss



EEG and Brain Waves

- Electroencephalograph (EEG): brain-wave machine; amplifies and records electrical activity in the brain
- Beta waves: small, fast waves associated with alertness and wakefulness
- Alpha waves: larger, slower waves associated with relaxation and falling asleep
- Delta waves: largest, slowest waves associated with deep sleep



100% Sleep Cycle

Stage 1

Stage 2

Stage 3

Stage 4

Dreaming

4-5%

Light sleep. Muscle activity slows down. Occasional muscle twitching.

45-55%

Breathing pattern and heart rate slows. Slight decrease in body temperature.

4-6%

Deep sleep begins. Brain begins to generate slow delta waves.

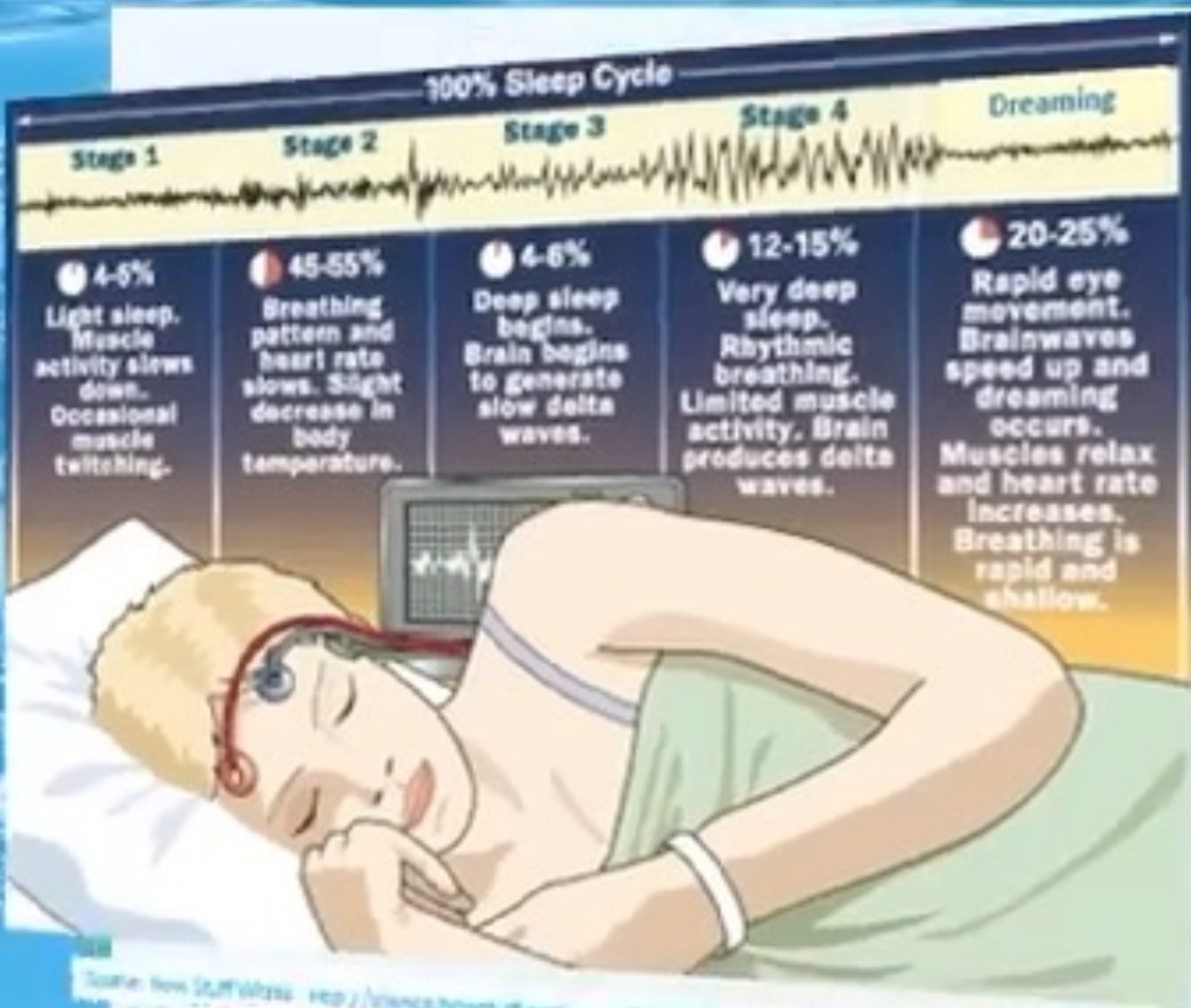
12-15%

Very deep sleep. Rhythmic breathing. Limited muscle activity. Brain produces delta waves.

20-25%

Rapid eye movement. Brain waves speed up and dreaming occurs. Muscles relax and heart rate increases. Breathing is rapid and shallow.





Source: New Staff Writer <http://science.howstuffworks.com/446/inside-the-mind/human-brain/dream2.htm>

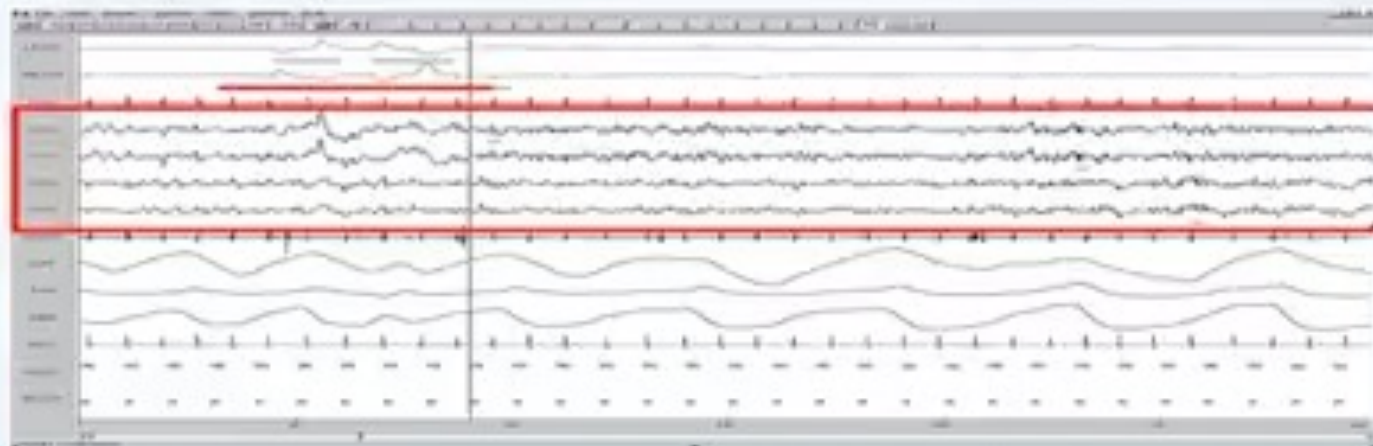
Dual-Process Hypothesis

- Two basic states of sleep are REM sleep and non-REM sleep.
- Dual-process in which sleep begins with non-REM sleep and then REM sleep.

REM sleep is characterized by rapid eye movement and is the stage of sleep in which most dreaming occurs.

Dual-Process Hypothesis of Sleep

- Two basic states of sleep are *non-REM (NREM)* sleep and *REM (rapid eye movement)* sleep
- Dual-process hypothesis of sleep: REM & NREM sleep help “refresh” the brain & store memories



REM sleep; EEG highlighted by red box; eye movements highlighted by red line. Thirty seconds of sleep.
Source Wikipedia Image Sleep



Function of REM Sleep



- Associated with dreaming; sleep is very light
- Return to Stage 1 sleep EEG patterns
- Body is very still during REM sleep – we are semi paralyzed
- REM sleep appears to “sharpen” our memories of the previous day’s more important experiences
- REM rebound: occurrence of extra REM sleep following REM sleep deprivation

Sleep Disturbances: Insomnia

- Insomnia: difficulty in getting to sleep, frequent nighttime awakenings, or waking too early
- Temporary insomnia: brief period of sleeplessness caused by worry, stress, and excitement
 - Avoid fighting it; get up and do something satisfying; return to bed when struggling to stay awake



Sleep Disturbances: Sleepwalking, Sleptalking, and Sleepsex

- Sleepwalking (somnambulism): occurs in NREM sleep during Stages 3 and 4
- Sleptalking: speaking while asleep; occurs in NREM sleep
- Many other behaviors, such as sleepsex, can also occur during NREM sleep



Sleep Disturbances: Nightmares and Night Terrors

- Nightmares: bad dreams that occur during REM sleep
- Night terrors: total panic and hallucinations may occur during Stage 4 NREM sleep
 - Most common in childhood; may occur in adults



Sleep Disturbances: Sleep Apnea

- Sleep apnea: repeated interruption of breathing during sleep
- Produces loud snoring with short silences and gasps for breath
- Apnea can be treated by
 - surgery
 - weight loss
 - continuous positive airway pressure (CPAP) mask



Sleep Disturbances: Sudden Infant Death Syndrome

- Sudden Infant Death Syndrome (SIDS; crib death): sudden, unexplained death of healthy infant
 - May be related to sleep apnea
 - May have weak arousal reflex
 - Infants should sleep on back to try to prevent it; remember “back to sleep”



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• Sudden Infant Death Syndrome (SIDS) is the sudden, unexplained death of a healthy infant, usually between the ages of 1 month and 1 year.

• SIDS is a leading cause of death for infants in the United States.

• The exact cause of SIDS is unknown, but it is thought to be related to a combination of factors, including sleep apnea, weak arousal reflex, and environmental factors.

• To reduce the risk of SIDS, it is recommended that infants sleep on their backs, in a crib with a firm mattress, and without loose bedding or toys.

• It is also important to avoid smoking during pregnancy and to avoid exposure to secondhand smoke.



Reality of Hypnosis



- People must cooperate to be hypnotized
- Basic suggestion effect: tendency of hypnotized people to carry out suggested actions as though they were involuntary
- Hypnotic susceptibility: how easily a person can be hypnotized

Hypnosis Can...

- Increase memory, both true and false
- Produce sensory changes
- Reduce pain



Hypnosis Cannot...

- Produce acts of superhuman strength
- Produce true age regression
- Force you to do things against your will



Stage Hypnosis

- Stage hypnosis: simulation of hypnotic effects
- Stage hypnotists entertain; they rarely hypnotize
- Waking suggestibility: people on stage do not want to spoil the act, so they will follow any instruction
- Selection of responsive subjects: any “volunteer” who does not get hypnotized in the group on stage and does not follow instructions is “thrown off”



Drug-Altered Consciousness

- Psychoactive drug: substance capable of altering attention, judgment, memory, time sense, self-control, emotion, or perception
- Stimulant: substance that increases activity in body and nervous system
- Depressant: substance that decreases activity in body and nervous system



Drug Dependence

- Physical dependence (addiction): compulsive use of a drug to maintain bodily comfort
 - Withdrawal symptoms: physical illness and discomfort following withdrawal of a drug
 - Drug tolerance: reduction in body's response to a drug
- Psychological dependence: use of drug necessary to maintain comfort or well-being
 - Crave drug and its rewarding qualities



Drug Issues

Drug interactions: when mixed, one drug enhances the effect of another; responsible for thousands of fatal drug overdoses every year

Drug tolerance: when the body needs more of the drug to gain the same effect from a lower dose





- Most frequently used psychoactive drug in North America; present in colas, chocolate, coffee, tea
- Causes tremors, sweating, talkativeness, tinnitus; suppresses fatigue or sleepiness, increases alertness
- Caffeinism: physiological dependence on caffeine
- Symptoms: insomnia, irritability, loss of appetite, chills, racing heart, elevated body temperature
- May be hazardous to pregnant women if used excessively; may cause birth defects

The End

